Kevin Eichner de Lemos Lisboa

AERODYNAMICS AND SIMULATION ENGINEER

k.lisboa@web.de Berlin linkedin.com/in/eichner-lisboa eichnerlisboa.github.io

ABOUT ME

As an Aerodynamics Engineer, I combine in-depth Computational Fluid Dynamics (CFD) with wind tunnel validation to achieve measurable performance gains. Through my leadership experience in Formula Student and a methodically rigorous Master's thesis, I possess a deep understanding of complex aerodynamic systems and their translation into high-performance applications.

SKILLS

- CFD: STAR-CCM+, Ansys Fluent
- CAD: NX, Solid Works
- Coding: Python, Matlab
- Experimental aerodynamics
- Project planning
- Data Analyze & Reporting
- AI Development & Application
- Team player
- Reliable
- Responsible
- Microsoft Office
- Driving license class B

LANGUAGE

- German (Native Speaker)
- English (fluent in writing and speech)

INTERESTS

- Sports: Climbing, Golf, Ski, Badminton
- DIY electronics/coding Projects

EDUCATION

Aeronautics and Astronautics, M.Sc.

TU Berlin 2021–Current preliminary. Grade: 1,5

Master's Thesis: "**A Combined Aerodynamic Investigation of Wind Tunnel Experiments and CFD Simulations of a Formula Student Race Car**" at the Chair of Fluid Dynamics supervised by Prof. Dr.-Ing. C.O. Paschereit, Dr.-Ing C. N. Nayeri

- **Methodology:** Performed a comprehensive correlation study between RANS simulations (STAR-CCM+) and physical wind tunnel data.
- **Scope:** Analyzed over **120 test cases and matching simulations**, evaluating various angles of attack, flow velocities, and vehicle configurations.

Mechanical engineering, B. Sc.

TU Berlin 2016–2021 Grade: 2,5

Thesis: „Entwicklung und Erprobung eines Versuchsstandes zur Analyse des Luftwiderstandes eines Röhrentransportsystems“ at the Chair of Fluid Dynamics supervised by Prof. Dr.-Ing. C.O. Paschereit, Dr.-Ing C. N. Nayeri

Abitur

Paulsen-Gymnasium 2010–2016 Grade: 2,2

LK: Math (Grade: 2), Physic (Grade: 2)

WORK EXPERIENCE

TU Berlin, Chair of Fluid Dynamics

04.2021-05.2025 Berlin Link

Student assistant

In this role, I supported academic research in the preparation, execution, and post-processing of fluid dynamics experiments.

- **Measurement Technology:** Utilized pressure and force sensors as well as Particle Image Velocimetry (PIV)
- **Data Acquisition:** Processed and analyzed raw experimental data using MATLAB and Python to support scientific publications
- **Design:** Developed and designed test rig components using **SolidWorks**

FaSTTUBe, Formula Student Team TU Berlin

09.2021-Aktuell Berlin Link

Head of Aerodynamics

09.2021-08.2022

In this leadership position, I was responsible for the strategic aerodynamic direction of the overall vehicle and the coordination of the entire department.

- **Leadership:** Led a multidisciplinary team of **15 members** (technical and personnel management)
- **Performance:** Achieved a **25% increase** in total downforce
- **Project Management:** Managed budget planning and ensured critical milestones from concept phase to final manufacturing

Head of Aero Performance

09.2022-08.2023

In this management role, I was responsible for the overall strategic and technical direction of the aerodynamics package as well as the technical leadership of the team.

- **Team Leadership:** Led **5 team members** and coordinated interdisciplinary interfaces (Suspension, Powertrain, Chassis)
- **Performance:** Achieved a **20% downforce increase** while reducing sensitivity to ride height variations
- **Development Loop:** Managed the end-to-end process from initial CAD sketching (**NX**) and CFD simulations (**STAR-CCM+**) to final design

Aero Design Engineer - Aerodynamic Performance

05.2019-08.2021, 09.2023-current

My focus was on the detailed design (**NX**) and layout (**STAR-CCM+**) of individual aerodynamic components and the acceleration of our digital development processes.

- **Development:** Focused on detailed design (**NX**) and analysis (**STAR-CCM+**) of aero-components
- **Design Success:** Secured **2nd Place in the Engineering Design Final** (FSAE Michigan, USA) by defending the concept before F1 and automotive industry experts
- **Automation:** Developed and optimized Python tools for workflow automation and data analysis

Freiberger Lebensmittel GmbH

07.2016-09.2016 **Berlin** **Link**

Internship in metal processing

This foundational internship provided me with the essential manual skills required for subsequent work on technical prototypes.

- **Manufacturing:** Acquired basic skills in turning, milling, drilling, and welding
- **Maintenance:** Assisted in the assembly and preventive maintenance of large-scale industrial systems